Story of How POSTECH Defeats KAIST

Mathematics for Science Quiz

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

POSTECH–KAIST Science War

Preface

This book is devoted to the Mathematics section of Science Quiz in the POSTECH–KAIST Science War. It collects mathematical concepts, facts to memorize, past problems, and expected problems that are useful for competition preparation.

The main purpose of this text is not merely to record solutions, but to build a practical and reliable mathematical toolkit for Science Quiz. It includes standard theorems, important terminology, memorable facts about mathematicians and mathematical history, and problem-solving strategies that may be useful in an actual match.

I sincerely hope that this book contributes to POSTECH's victory. Good luck, and let the game begin.

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Notation and Conventions

Throughout this book, we use standard mathematical notation whenever possible. The conventions below are arranged to keep the exposition readable, unified, and consistent with the chapter structure of Part I.

General Conventions

Scalars are written in ordinary italic letters, such as a, b, c, λ, t. Vectors are written in bold lowercase letters, such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Matrices are written in bold uppercase letters, such as $\mathbf{A}, \mathbf{B}, \mathbf{P}$. Unless otherwise stated, vectors are regarded as column vectors.

Notation	Meaning
$\mathbb{N}$	the set of positive integers, $\{1, 2, 3, \dots\}$
$\mathbb{N}_0$	the set of nonnegative integers, $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	the set of integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{R}$	the set of real numbers
$\mathbb{C}$	the set of complex numbers
$A \subseteq B$	A is a subset of B
$A \subsetneq B$	A is a proper subset of B
$A \setminus B$	the set difference of A and B
$\lvert A \rvert$	the cardinality of a finite set A
$\varnothing$	the empty set
λ	usually a scalar, eigenvalue, or parameter, depending on context
ε	a small positive real number, especially in analysis
$\sim$	may mean asymptotic equivalence, distribution, or an equivalence relation
$O(\cdot)$	Big-O notation
$o(\cdot)$	little-o notation
$\square$	end of a proof or solution

Calculus and Vector Calculus

Calculus notation includes one-variable calculus, multivariable calculus, and vector calculus. The symbol ∇ is read as “nabla” or “del.”

Notation	Meaning
$f'(a)$	the derivative of f at a
$f^{(n)}(a)$	the nth derivative of f at a

Notation	Meaning
$\dfrac{dy}{dx}$	the derivative of y with respect to x
$\displaystyle\int_a^b f(x)\,dx$	the definite integral of f from a to b
$C^k(I)$	the set of k times continuously differentiable functions on I
$C^\infty(I)$	the set of smooth functions on I
∇f or $\operatorname{grad} f$	the gradient of a scalar-valued function f
$\nabla \cdot \mathbf{F}$ or $\operatorname{div}\mathbf{F}$	the divergence of a vector field $\mathbf{F}$
$\nabla \times \mathbf{F}$ or $\operatorname{curl}\mathbf{F}$	the curl of a vector field $\mathbf{F}$
Δf	the Laplacian of f, usually $\nabla \cdot \nabla f$
$D\mathbf{F}$	the derivative matrix, or Jacobian matrix, of a map $\mathbf{F}$
$\operatorname{Jac}(\mathbf{F})$	the Jacobian determinant of $\mathbf{F}$, when defined
$\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r}$	a line integral of a vector field along a curve C
$\displaystyle\iint_S \mathbf{F}\cdot\mathbf{n}\,dS$	a flux integral across a surface S

Linear Algebra

In linear algebra, we consistently use bold notation for vectors and matrices. For example, $\mathbf{v} \in \mathbb{R}^n$ is a vector and $\mathbf{A} \in \mathbb{R}^{m\times n}$ is a matrix. The scalar λ is an eigenvalue, not a vector.

Notation	Meaning
$\mathbf{A}^\top$	the transpose of $\mathbf{A}$
$\mathbf{I}_n$	the $n \times n$ identity matrix
$\mathbf{0}$	the zero vector or zero matrix, depending on context
$\det(\mathbf{A})$	the determinant of $\mathbf{A}$
$\operatorname{rk}(\mathbf{A})$	the rank of $\mathbf{A}$
$\operatorname{tr}(\mathbf{A})$	the trace of $\mathbf{A}$
$\operatorname{adj}(\mathbf{A})$	the adjugate of $\mathbf{A}$
$\operatorname{Col}(\mathbf{A})$	the column space of $\mathbf{A}$
$\operatorname{Null}(\mathbf{A})$	the null space of $\mathbf{A}$
$\operatorname{Col}(\mathbf{A}^\top)$	the row space of $\mathbf{A}$
$\operatorname{Row}(\mathbf{A})$	another notation for the row space, used only when convenient
$\operatorname{Ker}(T)$	the kernel of a linear map T
$\operatorname{Im}(T)$	the image of a linear map T
$\operatorname{span}(\mathbf{v}_1,\dots,\mathbf{v}_k)$	the span of vectors $\mathbf{v}_1,\dots,\mathbf{v}_k$
$\langle \mathbf{u},\mathbf{v}\rangle$	the inner product of $\mathbf{u}$ and $\mathbf{v}$
$\lVert\mathbf{v}\rVert$	the norm of $\mathbf{v}$
$\mathbf{u} \perp \mathbf{v}$	$\mathbf{u}$ and $\mathbf{v}$ are orthogonal
$U \oplus V$	the direct sum of subspaces U and V

The four fundamental subspaces of $\mathbf{A} \in \mathbb{R}^{m\times n}$ are written as

$$\operatorname{Col}(\mathbf{A}), \qquad \operatorname{Null}(\mathbf{A}), \qquad \operatorname{Col}(\mathbf{A}^\top), \qquad \operatorname{Null}(\mathbf{A}^\top).$$

We use the standard orthogonal decompositions

$$\mathbb{R}^m = \operatorname{Col}(\mathbf{A}) \oplus \operatorname{Null}(\mathbf{A}^\top), \qquad \mathbb{R}^n = \operatorname{Col}(\mathbf{A}^\top) \oplus \operatorname{Null}(\mathbf{A}).$$

For cofactors, we write A_{ij} for the scalar cofactor of the entry a_{ij} of $\mathbf{A}$. The cofactor matrix is denoted by

$$\mathbf{C} = (A_{ij}),$$

and the adjugate matrix is

$$\operatorname{adj}(\mathbf{A}) = \mathbf{C}^{\mathsf{T}}.$$

We avoid using $\mathbf{A}^*$ for the adjugate, since $\mathbf{A}^*$ often denotes the conjugate transpose in other contexts.

Differential Equations

Differential equations are written using standard derivative notation. The independent variable is usually t or x, depending on the context.

Notation	Meaning
y'	the first derivative of y with respect to the relevant variable
y''	the second derivative of y with respect to the relevant variable
$\dot{x}$	the derivative of x with respect to time t
$\ddot{x}$	the second derivative of x with respect to time t
y_h	the homogeneous part of a solution
y_p	a particular solution
$c_1, c_2, \ldots$	arbitrary constants in a general solution
$\mathcal{L}\{f(t)\}(s)$	the Laplace transform of $f(t)$

Combinatorics, Discrete Mathematics, Probability and Statistics

Combinatorics and discrete mathematics provide the notation for counting, recurrence relations, and graphs. Probability and statistics use this notation frequently, especially in finite probability spaces.

Notation	Meaning
$n!$	factorial
$\binom{n}{k}$	the number of k-element subsets of an n-element set
$P(n,k)$	the number of k-permutations of an n-element set
a_n	the nth term of a sequence
$G = (V,E)$	a graph with vertex set V and edge set E
$\deg(v)$	the degree of a vertex v in a graph
K_n	the complete graph on n vertices
C_n	the cycle graph on n vertices
P_n	the path graph on n vertices
$\mathbb{P}(A)$	the probability of an event A
$\mathbb{P}(A \mid B)$	the conditional probability of A given B
$\mathbb{E}[X]$	the expectation of a random variable X
$\operatorname{Var}(X)$	the variance of X
$\operatorname{Cov}(X,Y)$	the covariance of X and Y
$X \sim \operatorname{Bin}(n,p)$	X follows a binomial distribution with parameters n and p
$X \sim \operatorname{Poi}(\lambda)$	X follows a Poisson distribution with parameter λ

Notation	Meaning
$X \sim \mathrm{N}(\mu, \sigma^2)$	X follows a normal distribution with mean μ and variance σ^2

Random variables are usually denoted by uppercase letters such as X, Y, Z. Events are denoted by uppercase letters such as A, B, E. In graph-theoretic contexts, $\deg(v)$ denotes the degree of a vertex.

Geometry

We use standard geometric notation unless otherwise stated.

Notation	Meaning
$\triangle ABC$	the triangle with vertices A, B, C
$\angle ABC$	the angle with vertex B
$\overline{AB}$	the line segment joining A and B
AB	the length of the segment $\overline{AB}$, when no confusion is possible
$[ABC]$	the area of triangle ABC, when used in geometry problems
$\parallel$	parallel
$\perp$	perpendicular

Number Theory and Algebra

Unless otherwise stated, divisibility and congruence statements are made in $\mathbb{Z}$. In algebra, the notation is kept standard and lightweight, since Science Quiz rarely requires highly specialized algebraic formalism.

Notation	Meaning
$a \mid b$	a divides b
$a \nmid b$	a does not divide b
$\gcd(a, b)$	the greatest common divisor of a and b
$\operatorname{lcm}(a, b)$	the least common multiple of a and b
$a \equiv b \pmod{n}$	a is congruent to b modulo n
$\varphi(n)$	Euler's phi function, or Euler's totient function
$\operatorname{ord}_n(a)$	the order of a modulo n
p	usually a prime number, when used in number-theoretic contexts
G	usually a group
e	the identity element of a group, when the context is clear
$H \leq G$	H is a subgroup of G
$H \trianglelefteq G$	H is a normal subgroup of G
G/H	the quotient group of G by H
R	usually a ring
F or K	usually a field
$F[x]$	the polynomial ring over a field F
$\deg f$	the degree of a polynomial f
$\operatorname{Aut}(G)$	the automorphism group of G
$\operatorname{Gal}(L/K)$	the Galois group of a field extension L/K

The Chinese Remainder Theorem may be abbreviated as CRT after it is first introduced. In algebraic contexts, $\deg f$ denotes the degree of a polynomial f; in graph-theoretic contexts, $\deg(v)$ denotes the degree of a vertex.

Miscellaneous Topics

Some symbols have different meanings in different areas. In this book, the intended meaning should always be clear from context.

Notation	Meaning
$\lfloor x \rfloor$	the floor of x
$\lceil x \rceil$	the ceiling of x
$\{x\}$	the fractional part of x, usually $x - \lfloor x \rfloor$

Part I

Mathematical Concepts

Chapter 1

Calculus

This chapter collects quiz-friendly results from one-variable calculus, multivariable calculus, and vector calculus. The emphasis is on named theorems, memorable formulas, and geometric ideas that are easy to ask and useful to recognize quickly.

One-Variable Calculus

Formula 1.1. Standard Trigonometric and Exponential Limits
The following limits are fundamental:

$$\lim_{x\to 0}\frac{\sin x}{x}=1, \qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2},$$

$$\lim_{x\to 0}\frac{e^x-1}{x}=1, \qquad \lim_{x\to 0}\frac{\log(1+x)}{x}=1,$$

and

$$\lim_{x\to 0}(1+x)^{1/x}=e.$$

Strategy. These limits are often faster than L'Hopital's Rule. In a quiz, the expressions $\sin x$, $1-\cos x$, e^x-1, $\log(1+x)$, and $(1+x)^{1/x}$ are strong signals for these standard patterns.

Theorem 1.2. Intermediate Value Theorem
Let f be continuous on $[a,b]$. If N is any number between $f(a)$ and $f(b)$, then there exists $c\in[a,b]$ such that

$$f(c)=N.$$

Strategy. The Intermediate Value Theorem is the theorem behind many existence arguments. To show that an equation has a solution, define a continuous function and show that its values cross the desired level.

Theorem 1.3. Extreme Value Theorem
Let f be continuous on $[a,b]$. Then f attains both a maximum and a minimum on $[a,b]$.

Remark. The closed interval condition is important. A continuous function on an open interval need not attain a maximum or a minimum.

Theorem 1.4. Mean Value Theorem
Let f be continuous on $[a,b]$ and differentiable on (a,b). Then there exists $c\in(a,b)$ such that

$$f'(c)=\frac{f(b)-f(a)}{b-a}.$$

Idea. The Mean Value Theorem says that somewhere on the interval, the instantaneous rate of change equals the average rate of change.

Theorem 1.5. Taylor's Theorem
Suppose that f is $n+1$ times differentiable near a. Then

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x),$$

where, in Lagrange's form of the remainder,

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

for some c between a and x.

Strategy. Taylor expansion is one of the fastest tools for limits, approximations, and hidden series evaluations. Expressions involving e^x, $\sin x$, $\cos x$, $\log(1+x)$, or $\arctan x$ often invite Taylor expansion.

Formula 1.6. Maclaurin Series
The most important Maclaurin series are

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \qquad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \qquad (|x| < 1),$$

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \qquad (-1 < x \le 1),$$

and

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \qquad (|x| \le 1).$$

Formula 1.7. Euler's Formula
For every real number x,

$$e^{ix} = \cos x + i \sin x.$$

In particular,

$$e^{i\pi} + 1 = 0.$$

Remark. The identity $e^{i\pi} + 1 = 0$ is called Euler's identity. It connects the constants e, i, π, 1, and 0 in a single formula.

Formula 1.8. Gaussian Integral
The Gaussian integral is

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}.$$

Equivalently,

$$\int_{0}^{\infty} e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2}.$$

Remark. The Gaussian integral is one of the most famous integrals in calculus. It is closely connected to the normal distribution in probability.

Formula 1.9. Wallis Product
The Wallis product is

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots .$$

Definition 1.10. Gamma Function
For $x > 0$, the Gamma function is defined by

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t}\, dt.$$

Proposition 1.11. Gamma Function and Factorials
The Gamma function satisfies

$$\Gamma(x+1) = x\Gamma(x).$$

In particular, for every positive integer n,

$$\Gamma(n) = (n-1)!.$$

Also,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Formula 1.12. Stirling's Formula
As $n \to \infty$,

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

Remark. Stirling's formula is the standard approximation for large factorials. It often appears in counting, probability, and asymptotic estimation.

Multivariable Calculus and Vector Calculus

Definition 1.13. Gradient, Divergence, Curl, and Laplacian
For a scalar-valued function $f(x, y, z)$ and a vector field $\mathbf{F} = (P, Q, R)$, we write

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right),$$

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z},$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix},$$

and

$$\Delta f = \nabla \cdot \nabla f = f_{xx} + f_{yy} + f_{zz}.$$

Idea. The gradient points in the direction of steepest increase. Divergence measures source or sink behavior. Curl measures local rotation. The Laplacian measures how a value differs from its surrounding average.

Theorem 1.14. Clairaut's Theorem
If f has continuous second partial derivatives near a point, then the mixed partial derivatives agree at that point:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Definition 1.15. Jacobian
For a map

$$\mathbf{F}(u_1, \ldots, u_n) = (x_1(u_1, \ldots, u_n), \ldots, x_n(u_1, \ldots, u_n)),$$

the Jacobian matrix is

$$D\mathbf{F} = \left(\frac{\partial x_i}{\partial u_j}\right)_{i,j=1}^{n}.$$

Its determinant

$$\mathrm{Jac}(\mathbf{F}) = \det(D\mathbf{F})$$

is called the Jacobian determinant.

Theorem 1.16. Change of Variables Formula
Under suitable regularity and one-to-one assumptions,

$$\int_{\mathbf{F}(D)} f(\mathbf{x})\, d\mathbf{x} = \int_D f(\mathbf{F}(\mathbf{u}))\, |\det D\mathbf{F}(\mathbf{u})|\; d\mathbf{u}.$$

Remark. The Jacobian determinant is the local area or volume scaling factor of a change of variables.

Theorem 1.17. Lagrange Multipliers
Suppose that f has a local maximum or minimum subject to the constraint

$$g(x_1, \ldots, x_n) = 0$$

at a point where $\nabla g \neq \mathbf{0}$. Then there exists a scalar λ such that

$$\nabla f = \lambda \nabla g.$$

Idea. At a constrained optimum, the level surface of f is tangent to the constraint surface. Thus their normal vectors are parallel.

Theorem 1.18. Green's Theorem
Let C be a positively oriented, simple closed curve in the plane, and let D be the region enclosed by C. Then

$$\oint_C P\, dx + Q\, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Idea. Green's Theorem converts circulation along a boundary curve into curl over the region inside. It is the two-dimensional ancestor of Stokes' Theorem.

Theorem 1.19. Stokes' Theorem
Let S be an oriented smooth surface with positively oriented boundary curve ∂S. Then

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS.$$

Idea. Stokes' Theorem says that circulation around the boundary equals total curl through the surface. It turns a boundary integral into an interior integral.

Theorem 1.20. Divergence Theorem
Let E be a solid region in $\mathbb{R}^3$ with outward-oriented boundary surface ∂E. Then

$$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_E \nabla \cdot \mathbf{F}\, dV.$$

Idea. The Divergence Theorem says that outward flux through the boundary equals total source strength inside the region.

Remark. Green's Theorem, Stokes' Theorem, and the Divergence Theorem share the same philosophy: a boundary integral is equal to an interior integral. This is the geometric heart of vector calculus.

Chapter 2

Linear Algebra

This chapter focuses on named theorems, matrix invariants, and geometric principles that are especially likely to appear in quiz-style questions.

Theorem 2.1. Rank–Nullity Theorem
Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

$$\dim V = \dim \mathrm{Ker}(T) + \dim \mathrm{Im}(T).$$

For a matrix $\mathbf{A} \in F^{m\times n}$, this becomes

$$n = \mathrm{rk}(\mathbf{A}) + \dim \mathrm{Null}(\mathbf{A}).$$

Strategy. Rank–Nullity connects variables, pivots, and free variables. It is the linear algebra version of accounting: nothing disappears; dimensions are redistributed.

Theorem 2.2. Invertible Matrix Theorem
For a square matrix $\mathbf{A} \in F^{n\times n}$, the following statements are equivalent:

$$\mathbf{A} \text{ is invertible}, \qquad \det(\mathbf{A}) \neq 0, \qquad \mathrm{rk}(\mathbf{A}) = n,$$

$$\mathrm{Null}(\mathbf{A}) = \{\mathbf{0}\},$$

and

$$\mathbf{Ax} = \mathbf{b}$$

has a unique solution for every $\mathbf{b} \in F^n$.

Remark. This theorem is a dictionary. It translates between determinants, rank, null spaces, and solvability of linear systems.

Definition 2.3. Determinant
The determinant is a scalar associated to a square matrix $\mathbf{A} \in F^{n\times n}$, denoted by $\det(\mathbf{A})$. Geometrically, for real matrices, $|\det(\mathbf{A})|$ is the volume scaling factor of the linear transformation $\mathbf{x} \mapsto \mathbf{Ax}$.

Proposition 2.4. Basic Properties of Determinants
For square matrices $\mathbf{A}, \mathbf{B} \in F^{n\times n}$, we have

$$\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}),$$

$$\det(\mathbf{A}^{\mathsf{T}}) = \det(\mathbf{A}),$$

and, if $\mathbf{A}$ is invertible,

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

Formula 2.5. Determinant of a Triangular Matrix
If $\mathbf{A}$ is upper triangular or lower triangular with diagonal entries $a_{11}, a_{22}, \ldots, a_{nn}$, then

$$\det(\mathbf{A}) = a_{11}a_{22}\cdots a_{nn}.$$

Proposition 2.6. Trace, Determinant, and Eigenvalues
Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity. Then

$$\operatorname{tr}(\mathbf{A}) = \lambda_1 + \cdots + \lambda_n$$

and

$$\det(\mathbf{A}) = \lambda_1\lambda_2\cdots\lambda_n.$$

Definition 2.7. Eigenvalue and Characteristic Polynomial
Let $\mathbf{A} \in F^{n\times n}$. A scalar $\lambda \in F$ is an eigenvalue of $\mathbf{A}$ if there exists a nonzero vector $\mathbf{v} \in F^n$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

The characteristic polynomial of $\mathbf{A}$ is

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A}).$$

Proposition 2.8. Eigenvalues and the Characteristic Polynomial
A scalar λ is an eigenvalue of $\mathbf{A}$ if and only if

$$\det(\lambda\mathbf{I}_n - \mathbf{A}) = 0.$$

Equivalently, the eigenvalues of $\mathbf{A}$ are the roots of the characteristic polynomial $p_{\mathbf{A}}(\lambda)$.

Theorem 2.9. Cayley–Hamilton Theorem
Every square matrix satisfies its own characteristic polynomial. That is, if

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A}),$$

then

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}.$$

Remark. Cayley–Hamilton is a famous named theorem. It is especially useful for reducing high powers of a matrix to lower powers.

Theorem 2.10. Diagonalization Criterion
A matrix $\mathbf{A} \in F^{n\times n}$ is diagonalizable over F if and only if F^n has a basis consisting of eigenvectors of $\mathbf{A}$.

Corollary 2.11. Distinct Eigenvalues Imply Diagonalizability
If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues in F, then $\mathbf{A}$ is diagonalizable over F.

Strategy. Distinct eigenvalues are an easy sufficient condition for diagonalizability. They are not necessary: a matrix can be diagonalizable even when some eigenvalues are repeated.

Theorem 2.12. Spectral Theorem for Real Symmetric Matrices
If $\mathbf{A} \in \mathbb{R}^{n\times n}$ is symmetric, then all eigenvalues of $\mathbf{A}$ are real, and there exists an orthogonal matrix $\mathbf{Q}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}.$$

Equivalently, $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$.

Remark. In quiz settings, the words "real symmetric matrix" are a strong signal for real eigenvalues and orthogonal diagonalization.

Theorem 2.13. Cauchy–Schwarz Inequality
For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$*, we have*

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|.$$

Equivalently, for real numbers $a_1, \ldots, a_n$ *and* $b_1, \ldots, b_n$*,*

$$\left(\sum_{i=1}^{n} a_i b_i \right)^2 \leq \left(\sum_{i=1}^{n} a_i^2 \right) \left(\sum_{i=1}^{n} b_i^2 \right).$$

Strategy. The expression $\sum a_i b_i$ is a strong signal for Cauchy–Schwarz. The vector form is often the fastest way to recognize the inequality.

Corollary 2.14. Triangle Inequality
For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$*, we have*

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Theorem 2.15. Gram–Schmidt Process
From any linearly independent list

$$\mathbf{v}_1, \ldots, \mathbf{v}_k$$

in an inner product space, one can construct an orthonormal list

$$\mathbf{q}_1, \ldots, \mathbf{q}_k$$

with the same span:

$$\operatorname{span}(\mathbf{q}_1, \ldots, \mathbf{q}_k) = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k).$$

Remark. Gram–Schmidt is the standard procedure for turning an arbitrary basis into an orthonormal basis.

Formula 2.16. Projection Matrix
If the columns of $\mathbf{Q} \in \mathbb{R}^{n \times k}$ are orthonormal, then the orthogonal projection matrix onto $\mathrm{Col}(\mathbf{Q})$ is

$$\mathbf{P} = \mathbf{Q}\mathbf{Q}^\mathsf{T}.$$

More generally, if the columns of $\mathbf{A} \in \mathbb{R}^{n \times k}$ are linearly independent, then the orthogonal projection matrix onto $\mathrm{Col}(\mathbf{A})$ is

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}.$$

Proposition 2.17. Projection Matrix Criterion
A matrix $\mathbf{P}$ *is a projection matrix if*

$$\mathbf{P}^2 = \mathbf{P}.$$

It is an orthogonal projection matrix if, in addition,

$$\mathbf{P}^\mathsf{T} = \mathbf{P}.$$

Definition 2.18. Singular Values
Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The singular values of $\mathbf{A}$ are the nonnegative square roots of the eigenvalues of

$$\mathbf{A}^\mathsf{T}\mathbf{A}.$$

Theorem 2.19. Singular Value Decomposition
Every real matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ *can be written as*

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\mathsf{T},$$

where $\mathbf{U}$ *and* $\mathbf{V}$ *are orthogonal matrices and* $\boldsymbol{\Sigma}$ *is a rectangular diagonal matrix whose diagonal entries are the singular values of* $\mathbf{A}$*.*

Remark. Singular Value Decomposition, or SVD, is one of the most powerful decompositions in linear algebra. It applies to every real matrix, not only to square or diagonalizable matrices.

Theorem 2.20. Jordan Canonical Form
Over the complex numbers, every square matrix is similar to a Jordan matrix. That is, for every $\mathbf{A} \in \mathbb{C}^{n\times n}$*, there exists an invertible matrix* $\mathbf{P}$ *such that*

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

is in Jordan canonical form.

Remark. Jordan canonical form is usually more important as a named concept than as a computational tool in Science Quiz. It describes what remains when a matrix cannot be fully diagonalized.

Chapter 3

Differential Equations

This chapter collects named principles and famous model equations from ordinary and partial differential equations.

Theorem 3.1. Picard–Lindelof Existence and Uniqueness Theorem
Consider the initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

If f and $\partial f/\partial y$ are continuous near (t_0, y_0), then there exists a unique solution near $t = t_0$.

Remark. In a quiz setting, the important phrase is "existence and uniqueness." It tells us when an initial condition determines exactly one solution.

Theorem 3.2. Superposition Principle
If $y_1, \dots, y_k$ are solutions of a homogeneous linear differential equation, then any linear combination

$$c_1 y_1 + \cdots + c_k y_k$$

is also a solution.

Remark. The Superposition Principle is the differential-equation version of linear algebra. It is one of the main reasons linear equations are much easier to handle than nonlinear equations.

Proposition 3.3. General Solution of a Nonhomogeneous Linear Equation
For a nonhomogeneous linear differential equation, the general solution has the form

$$y = y_h + y_p,$$

where y_h is the general solution of the associated homogeneous equation and y_p is one particular solution of the nonhomogeneous equation.

Formula 3.4. Second-Order Constant-Coefficient Equation
Consider the homogeneous equation

$$ay'' + by' + cy = 0, \qquad a \neq 0.$$

Its characteristic equation is

$$ar^2 + br + c = 0.$$

If the roots are distinct real numbers r_1, r_2, then

$$y_h = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

If the equation has a repeated real root r, then

$$y_h = (c_1 + c_2 t)e^{rt}.$$

If the roots are $\alpha \pm i\beta$ with $\beta \neq 0$, then

$$y_h = e^{\alpha t}\left(c_1 \cos \beta t + c_2 \sin \beta t\right).$$

Remark. The characteristic equation converts a constant-coefficient linear ODE into an algebraic equation. This is the key trick.

Definition 3.5. Wronskian
For two differentiable functions y_1 and y_2, their Wronskian is

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t).$$

Proposition 3.6. Wronskian and Linear Independence
If

$$W(y_1, y_2)(t_0) \neq 0$$

for some t_0, then y_1 and y_2 are linearly independent.

Formula 3.7. Exponential Growth and Decay
The equation

$$y' = ky$$

has solution

$$y(t) = Ce^{kt}.$$

If $k > 0$, this models exponential growth. If $k < 0$, this models exponential decay.

Formula 3.8. Logistic Equation
The logistic equation is

$$y' = ky\left(1 - \frac{y}{K}\right),$$

where $k > 0$ is the growth rate and $K > 0$ is the carrying capacity. Its equilibrium solutions are

$$y = 0 \qquad \text{and} \qquad y = K.$$

Remark. The logistic equation is the standard model for population growth with limited resources. The parameter K represents the limiting population.

Formula 3.9. Simple Harmonic Oscillator
The equation

$$x'' + \omega^2 x = 0$$

has general solution

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t.$$

Equivalently,

$$x(t) = A\cos(\omega t - \phi)$$

for suitable constants A and ϕ.

Remark. This is the equation behind ideal mass–spring motion and many small oscillation models in physics.

Formula 3.10. Damped Harmonic Oscillator
The equation

$$mx'' + cx' + kx = 0$$

models a damped harmonic oscillator. Its characteristic equation is

$$mr^2 + cr + k = 0.$$

The motion is underdamped, critically damped, or overdamped according as

$$c^2 - 4mk < 0, \qquad c^2 - 4mk = 0, \qquad c^2 - 4mk > 0.$$

Definition 3.11. Laplace Transform
The Laplace transform of a function $f(t)$ is

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty e^{-st} f(t)\,dt.$$

Proposition 3.12. Basic Laplace Transforms
The most common Laplace transforms are

$$\mathcal{L}\{1\}(s) = \frac{1}{s}, \qquad \mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a},$$
$$\mathcal{L}\{\sin at\}(s) = \frac{a}{s^2+a^2}, \qquad \mathcal{L}\{\cos at\}(s) = \frac{s}{s^2+a^2}.$$

Remark. The Laplace transform turns many differential equations into algebraic equations. This is its main computational advantage.

Definition 3.13. Heat Equation
The heat equation is the partial differential equation

$$u_t = \alpha u_{xx}, \qquad \alpha > 0.$$

It models diffusion, especially the flow of heat in a one-dimensional medium.

Definition 3.14. Wave Equation
The one-dimensional wave equation is

$$u_{tt} = c^2 u_{xx}.$$

It models waves traveling with speed c.

Theorem 3.15. d'Alembert's Formula
The solution of the wave equation

$$u_{tt} = c^2 u_{xx}$$

on the real line with initial conditions

$$u(x,0) = f(x), \qquad u_t(x,0) = g(x),$$

is

$$u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds.$$

Remark. d'Alembert's formula shows that waves move along the characteristic lines $x - ct =$ constant and $x + ct =$ constant.

Definition 3.16. Laplace Equation
Laplace's equation is

$$\Delta u = 0.$$

In two variables, this means

$$u_{xx} + u_{yy} = 0.$$

A function satisfying Laplace's equation is called harmonic.

Definition 3.17. Poisson Equation
Poisson's equation is

$$\Delta u = f.$$

It is an inhomogeneous version of Laplace's equation.

Remark. The heat equation, wave equation, and Laplace equation are three of the most famous PDEs. A quiz problem may ask for their names, forms, or physical interpretations.

Chapter 4

Combinatorics and Discrete Mathematics

This chapter collects high-yield results from combinatorics and discrete mathematics. The emphasis is on named principles, memorable formulas, and results with clear quiz-style answers.

Formula 4.1. Binomial Theorem; Newton's Binomial Formula
For every nonnegative integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

In particular,

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

Remark. The coefficients $\binom{n}{k}$ are the entries of Pascal's triangle. This is one of the most recognizable bridges between algebra and counting.

Theorem 4.2. Pigeonhole Principle; Dirichlet's Box Principle
If $n+1$ objects are placed into n boxes, then at least one box contains at least two objects. More generally, if N objects are placed into k boxes, then at least one box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

Strategy. The Pigeonhole Principle is often hidden in problems asking us to prove that two objects must share a property. The hard part is usually choosing the right "boxes."

Theorem 4.3. Inclusion–Exclusion Principle; Sieve Formula
For finite sets $A_1, \ldots, A_n$,

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|.$$

Remark. Inclusion–Exclusion is the standard correction mechanism for overcounting. It is also historically connected to sieve methods.

Formula 4.4. Derangement Formula
Let D_n be the number of derangements of n objects, that is, permutations with no fixed points. Then

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Moreover,

$$D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

Remark. The approximation $D_n \approx n!/e$ is very memorable. In probability language, a random permutation has no fixed point with probability approximately $1/e$.

Formula 4.5. Catalan Numbers
The nth Catalan number is

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

It begins as

$$1,\ 1,\ 2,\ 5,\ 14,\ 42,\ 132, \ldots.$$

Remark. Catalan numbers count many different objects, including correct parenthesizations, Dyck paths, triangulations of a polygon, and rooted full binary trees. The number 42 is a classic giveaway.

Theorem 4.6. Cayley's Formula
The number of labeled trees on n vertices is

$$n^{n-2}.$$

Remark. Cayley's formula is a perfect Science Quiz result: the statement is short, the answer is surprising, and the name is famous.

Theorem 4.7. Burnside's Lemma; Cauchy–Frobenius Lemma
Suppose a finite group G acts on a finite set X. The number of orbits is

$$\frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|,$$

where

$$\operatorname{Fix}(g) = \{x \in X : g \cdot x = x\}.$$

Idea. Burnside's Lemma says that the number of essentially different objects is the average number of objects fixed by a symmetry.

Lemma 4.8. Handshaking Lemma
For a finite undirected graph $G = (V, E)$,

$$\sum_{v \in V} \deg(v) = 2|E|.$$

In particular, the number of vertices of odd degree is even.

Remark. The name comes from the idea that every handshake contributes 2 to the total count of hands shaken.

Theorem 4.9. Euler's Formula for Planar Graphs
If a connected planar graph has V vertices, E edges, and F faces, then

$$V - E + F = 2.$$

Remark. This is the graph-theoretic version of Euler's polyhedron formula. It is one of the most famous formulas involving planar graphs.

Theorem 4.10. Eulerian Trail Criterion
A connected finite graph has an Eulerian circuit if and only if every vertex has even degree. It has an Eulerian trail but not an Eulerian circuit if and only if exactly two vertices have odd degree.

Remark. This theorem is historically connected to Euler's solution of the Seven Bridges of Konigsberg problem.

Theorem 4.11. Hall's Marriage Theorem
Let $G = (X \cup Y, E)$ be a bipartite graph. There is a matching that covers every vertex of X if and only if, for every subset $S \subseteq X$,

$$|N(S)| \geq |S|,$$

where $N(S)$ is the set of neighbors of vertices in S.

Strategy. Hall's Marriage Theorem is the standard named theorem for proving that a perfect assignment or matching exists.

Theorem 4.12. Ramsey's Theorem
For any positive integers r and s, there exists an integer $R(r, s)$ such that every red-blue coloring of the edges of a sufficiently large complete graph contains either a red K_r or a blue K_s.

Formula 4.13. The Party Problem
The classical Ramsey number

$$R(3, 3) = 6$$

says that among any six people, there are either three mutual acquaintances or three mutual strangers.

Remark. Ramsey theory is the mathematics of unavoidable structure. Its slogan is that complete disorder is impossible in a large enough system.

Theorem 4.14. Erdos–Szekeres Theorem
Any sequence of

$$(r - 1)(s - 1) + 1$$

distinct real numbers contains either an increasing subsequence of length r or a decreasing subsequence of length s.

Remark. This is a beautiful pigeonhole-style result. It is often remembered as the monotone subsequence theorem.

Chapter 5

Probability and Statistics

This chapter focuses on probability results that are easy to ask in a quiz: named theorems, famous distributions, memorable approximations, and inequalities with clear meanings.

Theorem 5.1. Law of Total Probability
If $B_1, \ldots, B_n$ form a partition of the sample space and $\mathbb{P}(B_i) > 0$ for all i, then

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Theorem 5.2. Bayes' Theorem
If $B_1, \ldots, B_n$ form a partition of the sample space and $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

In the two-event form,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Remark. Bayes' Theorem is the standard formula for reversing conditional probability. It updates a prior probability after observing evidence.

Proposition 5.3. Linearity of Expectation
For random variables $X_1, \ldots, X_n$,

$$\mathbb{E}[X_1 + \cdots + X_n] = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n].$$

This holds even when the random variables are not independent.

Strategy. Linearity of expectation is often the fastest way to count an expected number of objects. Introduce indicator random variables and add their expectations.

Formula 5.4. Variance and Covariance Identities
For a random variable X,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

For random variables X and Y,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y).$$

If X and Y are independent, then

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Formula 5.5. Binomial Distribution
If $X \sim \mathrm{Bin}(n,p)$, then

$$\mathbb{P}(X=k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0,1,\ldots,n.$$

Its mean and variance are

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Formula 5.6. Poisson Distribution
If $X \sim \mathrm{Poi}(\lambda)$, then

$$\mathbb{P}(X=k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0,1,2,\ldots.$$

Its mean and variance are

$$\mathbb{E}[X] = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

Theorem 5.7. Poisson Limit Theorem; Law of Small Numbers
If $n \to \infty$, $p \to 0$, and $np \to \lambda$, then

$$\binom{n}{k} p^k (1-p)^{n-k} \to e^{-\lambda}\frac{\lambda^k}{k!}$$

for each fixed $k \geq 0$.

Remark. This theorem explains why the Poisson distribution models rare events: many trials, small success probability, and finite expected count.

Formula 5.8. Normal Distribution; Gaussian Distribution
A random variable $X \sim \mathrm{N}(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The standard normal distribution is $\mathrm{N}(0,1)$.

Formula 5.9. Empirical Rule; 68–95–99.7 Rule
For a normal distribution, approximately

$$68\%$$

of the mass lies within one standard deviation of the mean,

$$95\%$$

lies within two standard deviations, and

$$99.7\%$$

lies within three standard deviations.

Theorem 5.10. Weak Law of Large Numbers
Let $X_1, X_2, \ldots$ be independent identically distributed random variables with mean μ. Then, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{X_1 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right) \to 0$$

as $n \to \infty$.

Idea. The Law of Large Numbers says that the sample average stabilizes near the true mean when the number of trials becomes large.

Theorem 5.11. Central Limit Theorem
Let $X_1, X_2, \dots$ be independent identically distributed random variables with mean μ and variance $\sigma^2 > 0$. Then

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to the standard normal distribution $\mathrm{N}(0, 1)$.

Remark. The Central Limit Theorem explains why the normal distribution appears so often in nature and statistics.

Theorem 5.12. Markov's Inequality
If X is a nonnegative random variable and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 5.13. Chebyshev's Inequality
If X has mean μ and variance σ^2, then for every $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

Remark. Markov's inequality uses only the mean, while Chebyshev's inequality uses the variance. Both are probability bounds that work without assuming a normal distribution.

Theorem 5.14. Jensen's Inequality
If φ is convex, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

If φ is concave, the inequality is reversed.

Idea. Jensen's inequality says that, for a convex function, applying the function after averaging gives a value no larger than averaging after applying the function.

Formula 5.15. Birthday Problem Approximation
In a group of n people, assuming 365 equally likely birthdays, the probability that at least two people share a birthday is approximately

$$1 - \exp\left(-\frac{n(n-1)}{730}\right).$$

For $n = 23$, this probability is already greater than $1/2$.

Remark. The birthday problem is famous because the answer is surprisingly small: only 23 people are needed for a better-than-even chance of a shared birthday.

Formula 5.16. Gambler's Ruin
Suppose a gambler starts with i dollars and stops when reaching either 0 dollars or N dollars. If the probability of winning each round is p and $q = 1 - p$, then the probability of reaching N before 0 is

$$\frac{1 - (q/p)^i}{1 - (q/p)^N} \qquad (p \neq q).$$

In the fair case $p = q = 1/2$, the probability is

$$\frac{i}{N}.$$

Remark. Gambler's ruin is a classic random walk result. It is a good example where the fair case has a very simple answer.

Chapter 6

Geometry

Chapter 7

Number Theory

This chapter collects famous number-theoretic results with strong quiz value: primes, congruences, reciprocity, sums of squares, and the distribution of primes.

Theorem 7.1. Fundamental Theorem of Arithmetic
Every integer $n > 1$ can be written as a product of prime numbers. Moreover, this factorization is unique up to the order of the factors. Equivalently,

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where $p_1, \dots, p_k$ are distinct primes and $\alpha_1, \dots, \alpha_k$ are positive integers, and this expression is unique up to reordering.

Remark. This theorem is often called unique prime factorization. It is the reason prime numbers are treated as the basic building blocks of integers.

Theorem 7.2. Euclid's Theorem; Infinitude of Primes
There are infinitely many prime numbers.

Idea. If $p_1, \dots, p_n$ are finitely many primes, consider

$$p_1 p_2 \cdots p_n + 1.$$

This number is not divisible by any prime on the list.

Theorem 7.3. Bezout's Identity
Let $a, b \in \mathbb{Z}$, not both zero. Then there exist integers $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b).$$

In particular, if $\gcd(a, b) = 1$, then there exist integers $x, y \in \mathbb{Z}$ such that

$$ax + by = 1.$$

Remark. Bezout's identity is the algebraic heart of the Euclidean algorithm and modular inverses.

Theorem 7.4. Chinese Remainder Theorem; CRT
Let $m_1, \dots, m_k$ be pairwise relatively prime positive integers. Then the system

$$x \equiv a_1 \pmod{m_1}, \qquad x \equiv a_2 \pmod{m_2}, \qquad \dots, \qquad x \equiv a_k \pmod{m_k}$$

has a solution. Moreover, the solution is unique modulo

$$M = m_1 m_2 \cdots m_k.$$

Strategy. CRT is the standard theorem for combining several congruence conditions into one congruence.

Formula 7.5. Euler's Phi Function Formula
If

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

is the prime factorization of n, then

$$\varphi(n) = n \prod_{i=1}^{k} \left(1 - \frac{1}{p_i}\right).$$

In particular, for a prime power,

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1).$$

Theorem 7.6. Fermat's Little Theorem
Let p be a prime number. If $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for every integer a,

$$a^p \equiv a \pmod{p}.$$

Theorem 7.7. Euler's Theorem
If $\gcd(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Remark. Fermat's Little Theorem is the special case of Euler's Theorem when $n = p$ is prime, since $\varphi(p) = p - 1$.

Theorem 7.8. Wilson's Theorem
A positive integer $p > 1$ is prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

Remark. Wilson's theorem is famous because it gives a clean factorial characterization of primes, even though it is not efficient as a primality test.

Definition 7.9. Legendre Symbol
Let p be an odd prime. The Legendre symbol is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1, & \text{if } a \text{ is a quadratic nonresidue modulo } p, \\ 0, & \text{if } p \mid a. \end{cases}$$

Theorem 7.10. Euler's Criterion
Let p be an odd prime. For any integer a,

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Equivalently, if $p \nmid a$, then

$$a^{(p-1)/2} \equiv \begin{cases} 1 \pmod{p}, & \textit{if } a \textit{ is a quadratic residue modulo } p, \\ -1 \pmod{p}, & \textit{if } a \textit{ is a quadratic nonresidue modulo } p. \end{cases}$$

Theorem 7.11. Law of Quadratic Reciprocity; Gauss's Theorem of Quadratic Reciprocity
Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$$

unless both p and q are congruent to 3 modulo 4.

Remark. Quadratic reciprocity is one of the crown jewels of elementary number theory. Gauss called it the golden theorem.

Theorem 7.12. Fermat's Two-Square Theorem
An odd prime p can be written as a sum of two squares,

$$p = a^2 + b^2$$

for some integers $a, b \in \mathbb{Z}$, if and only if

$$p \equiv 1 \pmod 4.$$

Also,

$$2 = 1^2 + 1^2.$$

Theorem 7.13. Lagrange's Four-Square Theorem
Every nonnegative integer can be written as a sum of four integer squares. That is, for every $n \in \mathbb{N}_0$, there exist integers $a, b, c, d \in \mathbb{Z}$ such that

$$n = a^2 + b^2 + c^2 + d^2.$$

Remark. The two-square theorem has a congruence condition. The four-square theorem has no exception at all.

Theorem 7.14. Dirichlet's Theorem on Arithmetic Progressions
If a and d are relatively prime positive integers, then the arithmetic progression

$$a,\ a+d,\ a+2d,\ a+3d, \dots$$

contains infinitely many primes.

Remark. Dirichlet's theorem is a major result connecting primes and arithmetic progressions. The condition $\gcd(a, d) = 1$ is necessary.

Theorem 7.15. Bertrand's Postulate; Chebyshev's Theorem
For every integer $n > 1$, there exists a prime p such that

$$n < p < 2n.$$

Theorem 7.16. Prime Number Theorem
Let $\pi(x)$ be the number of primes less than or equal to x. Then

$$\pi(x) \sim \frac{x}{\log x} \qquad \textit{as } x \to \infty.$$

Idea. The Prime Number Theorem says that the "probability" that a large integer near x is prime is roughly $1/\log x$. This is a mental model, not a literal probability statement.

Definition 7.17. Riemann Zeta Function
For $\mathrm{Re}(s) > 1$, the Riemann zeta function is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

It also has the Euler product

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

Remark. The Euler product is the bridge between the zeta function and prime numbers. The Riemann Hypothesis asserts that the nontrivial zeros of $\zeta(s)$ all have real part $1/2$.

Theorem 7.18. Euclid–Euler Theorem on Perfect Numbers
An even positive integer N is perfect if and only if

$$N = 2^{p-1}(2^p - 1),$$

where $2^p - 1$ is a Mersenne prime.

Remark. A perfect number is a positive integer equal to the sum of its positive proper divisors. For example,

$$6 = 1 + 2 + 3.$$

Theorem 7.19. Fermat's Last Theorem
For every integer $n \geq 3$, there do not exist positive integers a, b, c such that

$$a^n + b^n = c^n.$$

Remark. Fermat's Last Theorem was proved by Andrew Wiles. It is one of the most famous theorems in the history of mathematics.

Chapter 8

Algebra

Chapter 9

Analysis

Chapter 10

Topology

Chapter 11

Miscellaneous Topics

Classical Infinite Series

Formula 11.1. Mercator Series for $\ln 2$
The alternating harmonic series satisfies

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2.$$

Remark. This follows from the Mercator series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

by substituting $x = 1$.

Formula 11.2. Basel Sum; Euler's Basel Formula
The Basel sum is

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}.$$

Equivalently,

$$\zeta(2) = \frac{\pi^2}{6}.$$

Remark. The problem of evaluating

$$\sum_{n=1}^{\infty}\frac{1}{n^2}$$

is called the Basel problem. Its famous solution was found by Euler.

Formula 11.3. Odd Part of the Basel Sum
The sum of reciprocal squares over odd positive integers is

$$\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}.$$

Idea. This is obtained by subtracting the even part from the Basel sum:

$$\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \sum_{n=1}^{\infty}\frac{1}{n^2} - \sum_{n=1}^{\infty}\frac{1}{(2n)^2} = \left(1 - \frac{1}{4}\right)\frac{\pi^2}{6} = \frac{\pi^2}{8}.$$

Formula 11.4. Alternating Basel Series; Dirichlet Eta Value at 2
The alternating Basel series satisfies

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}.$$

Equivalently,

$$\eta(2) = \frac{\pi^2}{12},$$

where $\eta(s)$ is the Dirichlet eta function.

Idea. The positive terms are the odd reciprocal squares and the negative terms are the even reciprocal squares. Hence

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^2} = \sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} - \sum_{n=1}^{\infty}\frac{1}{(2n)^2} = \frac{\pi^2}{8} - \frac{\pi^2}{24} = \frac{\pi^2}{12}.$$

Formula 11.5. Gregory–Leibniz Series; Leibniz Formula for π
The Gregory–Leibniz series is

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{2n-1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

Equivalently,

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right).$$

Remark. This follows from the Taylor series

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

by substituting $x = 1$, since $\arctan 1 = \pi/4$.

Strategy. These values are useful because Science Quiz problems often hide them behind a simple transformation. For example,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right) = \frac{\ln 2}{2}.$$

Part II

Facts to Memorize

Chapter 12

ICM and the Fields Medal

The International Congress of Mathematicians, commonly abbreviated as ICM, is the most important international congress in mathematics. The Fields Medal is awarded by the International Mathematical Union at the ICM every four years. A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress.

Year	ICM location	Medalists and areas	Notes
1936	Oslo, Norway	Lars Ahlfors — complex analysis, Riemann surfaces; Jesse Douglas — minimal surfaces, Plateau problem	First Fields Medals.
1950	Cambridge, USA	Laurent Schwartz — distribution theory; Atle Selberg — analytic number theory, sieve methods, zeta functions	First awards after World War II.
1954	Amsterdam, Netherlands	Kunihiko Kodaira — complex geometry, algebraic geometry; Jean-Pierre Serre — algebraic topology, sheaf theory	Serre is the youngest Fields Medalist.
1958	Edinburgh, UK	Klaus Roth — number theory, Diophantine approximation; Rene Thom — cobordism, differential topology	
1962	Stockholm, Sweden	Lars Hormander — partial differential equations; John Milnor — differential topology, exotic spheres	
1966	Moscow, USSR	Michael Atiyah — K-theory, index theorem; Paul Cohen — set theory, forcing, continuum hypothesis; Alexander Grothendieck — algebraic geometry, homological algebra; Stephen Smale — differential topology, generalized Poincare conjecture	First year with four medalists.
1970	Nice, France	Alan Baker — transcendental number theory; Heisuke Hironaka — algebraic geometry, resolution of singularities; Serge Novikov — topology, differential geometry; John G. Thompson — finite group theory	
1974	Vancouver, Canada	Enrico Bombieri — number theory, complex analysis, minimal surfaces; David Mumford — algebraic geometry, moduli spaces	
1978	Helsinki, Finland	Pierre Deligne — algebraic geometry, Weil conjectures; Charles Fefferman — analysis, several complex variables; Grigory Margulis — Lie groups, ergodic theory, discrete groups; Daniel Quillen — algebraic K-theory	

Continued on the next page.

Year	ICM location	Medalists and areas	Notes
1982	Warsaw, Poland	Alain Connes — operator algebras, noncommutative geometry; William Thurston — low-dimensional topology, 3-manifolds; Shing-Tung Yau — differential geometry, PDE, Calabi conjecture	ICM Warsaw was held in 1983.
1986	Berkeley, USA	Simon Donaldson — four-manifold topology, gauge theory; Gerd Faltings — arithmetic algebraic geometry, Mordell conjecture; Michael Freedman — four-manifold topology, four-dimensional Poincare conjecture	
1990	Kyoto, Japan	Vladimir Drinfeld — quantum groups, Langlands program; Vaughan Jones — operator algebras, knot theory; Shigefumi Mori — algebraic geometry, minimal model program; Edward Witten — mathematical physics, topology, quantum field theory	Witten was the first physicist to receive the Fields Medal.
1994	Zurich, Switzerland	Jean Bourgain — analysis, Banach spaces, harmonic analysis; Pierre-Louis Lions — partial differential equations; Jean-Christophe Yoccoz — dynamical systems; Efim Zelmanov — algebra, restricted Burnside problem	
1998	Berlin, Germany	Richard Borcherds — algebra, automorphic forms, moonshine; W. Timothy Gowers — functional analysis, combinatorics; Maxim Kontsevich — algebraic geometry, topology, mathematical physics; Curtis T. McMullen — complex dynamics, hyperbolic geometry	Andrew Wiles received a special IMU silver plaque.
2002	Beijing, China	Laurent Lafforgue — Langlands correspondence over function fields; Vladimir Voevodsky — motivic cohomology, $\mathbb{A}^1$-homotopy theory	
2006	Madrid, Spain	Andrei Okounkov — probability, representation theory, algebraic geometry; Grigori Perelman — Ricci flow, Poincare conjecture; Terence Tao — PDE, harmonic analysis, additive combinatorics; Wendelin Werner — probability, SLE, Brownian motion	Perelman declined the medal.
2010	Hyderabad, India	Elon Lindenstrauss — ergodic theory, number theory; Ngo Bao Chau — automorphic forms, Fundamental Lemma; Stanislav Smirnov — probability, statistical physics, conformal invariance; Cedric Villani — kinetic theory, Boltzmann equation, optimal transport	
2014	Seoul, South Korea	Artur Avila — dynamical systems; Manjul Bhargava — number theory, geometry of numbers; Martin Hairer — stochastic partial differential equations; Maryam Mirzakhani — Riemann surfaces, moduli spaces, dynamics	Mirzakhani was the first woman Fields Medalist.
2018	Rio de Janeiro, Brazil	Caucher Birkar — algebraic geometry, Fano varieties, minimal model program; Alessio Figalli — optimal transport, PDE, metric geometry; Peter Scholze — arithmetic algebraic geometry, p-adic geometry; Akshay Venkatesh — number theory, homogeneous dynamics, representation theory	

Continued on the next page.

Year	ICM location	Medalists and areas	Notes
2022	Helsinki, Finland	Hugo Duminil-Copin — probability, statistical physics, phase transitions; June Huh — Hodge theory, combinatorics, matroids; James Maynard — analytic number theory, prime numbers; Maryna Viazovska — sphere packing, E_8 lattice, Fourier analysis	ICM 2022 was virtual; the award ceremony was held in Helsinki. Viazovska was the second woman Fields Medalist.

Essential ICM and Fields Medal Landmarks

- ICM stands for International Congress of Mathematicians.
- IMU stands for International Mathematical Union.
- The Fields Medal is awarded every four years at the ICM.
- The Fields Medal is named after John Charles Fields.
- The Fields Medal was first awarded in 1936.
- The first two Fields Medalists were Lars Ahlfors and Jesse Douglas.
- A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress.
- Jean-Pierre Serre is the youngest Fields Medalist.
- Edward Witten was the first physicist to receive the Fields Medal.
- Andrew Wiles received a special IMU silver plaque in 1998 for his proof of Fermat's Last Theorem.
- Grigori Perelman declined the Fields Medal in 2006 after proving the Poincare conjecture.
- Maryam Mirzakhani was the first woman Fields Medalist.
- Maryna Viazovska was the second woman Fields Medalist.
- The 2026 ICM is scheduled to be held in Philadelphia, USA.

Part III

Past Problems

Problem 2018.1. The following are this year's Fields Medalists. Excluding choice (5), who is the second oldest among them?

(1) Caucher Birkar
(2) Alessio Figalli
(3) Peter Scholze
(4) Akshay Venkatesh
(5) The person who stole Birkar's medal

Solution. The 2018 Fields Medalists were

(1)	Caucher Birkar	born in 1978,
(2)	Alessio Figalli	born in 1984,
(3)	Peter Scholze	born in 1987,
(4)	Akshay Venkatesh	born in 1981.

Among these four, the second oldest is Akshay Venkatesh. Therefore, the answer is

$$\boxed{(4)}.$$

Choice (5) is a joke referring to the incident in which Caucher Birkar's Fields Medal was stolen shortly after the 2018 Fields Medal ceremony. Birkar later received a replacement medal. □

Problem 2018.2. According to William Thurston's geometrization conjecture, a 3-dimensional manifold can be cut along spheres and tori so that each resulting piece has one of n possible geometric structures. What is n?

Solution. The 3-dimensional geometric structures appearing in Thurston's geometrization conjecture are the eight Thurston geometries:

$$S^3, \quad \mathbb{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}_2(\mathbb{R})}, \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Hence $n = 8$, so the answer is

$$\boxed{8}.$$

□

Problem 2018.3. Find the sum of the following infinite series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots .$$

Solution. The given series is one half of the alternating harmonic series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right).$$

Recall the Taylor expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots .$$

Substituting $x = 1$, we obtain

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots .$$

Therefore,

$$\frac{1}{2}-\frac{1}{4}+\frac{1}{6}-\frac{1}{8}+\cdots=\frac{\ln 2}{2}.$$

Thus the answer is $\boxed{\dfrac{\ln 2}{2}}$.

□

Problem 2018.4. Among the following five Fields Medalists, which one does not have a Korean doctoral student?

(1) William Thurston
(2) Maxim Kontsevich
(3) Grigory Margulis
(4) Terence Tao
(5) Efim Zelmanov

Solution. Maxim Kontsevich

□

Problem 2020.1. The International Congress of Mathematicians, abbreviated as ICM, is a congress for mathematicians from around the world. It is organized every four years by the International Mathematical Union and may be thought of as a kind of Olympic Games for mathematicians. The 2014 ICM was held in South Korea, and the 2018 ICM was held in Brazil. As of 2019, in which country was the 2022 ICM scheduled to be held?

Solution. ICM stands for the International Congress of Mathematicians. It is the main international congress in mathematics and is organized every four years by the International Mathematical Union.

The 2014 ICM was held in Seoul, South Korea, and the 2018 ICM was held in Rio de Janeiro, Brazil. At the time this problem was asked in 2019, the 2022 ICM was scheduled to be held in Saint Petersburg, Russia. Therefore, the answer is $\boxed{\text{Russia}}$.

In reality, after Russia's invasion of Ukraine in 2022, the 2022 ICM was not held in Saint Petersburg. It was eventually held as a virtual event, with the award ceremony taking place in Helsinki, Finland. Thus, the answer to this problem should be understood according to the planned host country at the time of the 2019 competition.

□

Problem 2020.2. A fair six-sided die has the numbers $-2, 0, 2, 3, 4, 5$ written on its faces. The die is rolled 10 times, and X is defined to be the product of the 10 numbers obtained. What is the expected value of X?

Solution. Let Y be the number obtained from one roll of the die. Since all six faces are equally likely,

$$\mathbb{E}[Y]=\frac{-2+0+2+3+4+5}{6}=2.$$

Now let $Y_1, Y_2, \ldots, Y_{10}$ be the numbers obtained from the 10 rolls. Then

$$X=Y_1Y_2\cdots Y_{10}.$$

Since the rolls are independent, the random variables $Y_1, Y_2, \ldots, Y_{10}$ are independent. Therefore,

$$\mathbb{E}[X]=\mathbb{E}[Y_1Y_2\cdots Y_{10}]=\mathbb{E}[Y_1]\mathbb{E}[Y_2]\cdots\mathbb{E}[Y_{10}].$$

All the Y_i have the same distribution as Y, so

$$\mathbb{E}[X] = (\mathbb{E}[Y])^{10} = 2^{10} = 1024.$$

Thus the answer is $\boxed{1024}$.

□

Problem 2020.3. Which of the following Fields Medalists was not serving as a professor at the time?

(1) Cédric Villani
(2) Elon Lindenstrauss
(3) Peter Scholze
(4) Martin Hairer
(5) Terence Tao

Solution. At the time this problem was asked, Cédric Villani was active as a politician in France rather than serving as a professor.

The other choices were mathematicians holding professorial positions. Therefore, the answer is $\boxed{\text{(1) Cédric Villani}}$.

□

Problem 2020.4. In a three-dimensional gravitational field, when an object moves from a point $\mathbf{a}$ to another point $\mathbf{b}$, the change in its potential energy is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$. Which of the following mathematical facts is least directly related to this physical phenomenon?

(1) A curl-free vector field is the gradient of some potential function.
(2) Stokes' theorem
(3) The Laplacian of $\dfrac{1}{|\mathbf{x}|}$ is a delta function, up to a constant factor.
(4) The curl of a gradient is identically zero.

Solution. The statement that the change in potential energy is independent of the path means that the gravitational field is a conservative vector field.

Let $\mathbf{F}$ be the force field. The change in potential energy from $\mathbf{a}$ to $\mathbf{b}$ is given by

$$\Delta U = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{r}.$$

For this quantity to be independent of the path, the force field should be expressible as the negative gradient of a potential function:

$$\mathbf{F} = -\nabla U.$$

Thus choice (1) is directly related to the phenomenon.

Also, the curl of a gradient is always zero:

$$\nabla \times \nabla U = \mathbf{0}.$$

Hence choice (4) is also directly related. Stokes' theorem says that

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dS.$$

Therefore, if $\nabla \times \mathbf{F} = \mathbf{0}$, then the line integral around a closed curve is zero, which is another way to understand path independence. Thus choice (2) is also related.

On the other hand,

$$\Delta\left(\frac{1}{|\mathbf{x}|}\right) = -4\pi\delta(\mathbf{x})$$

is related to the fundamental solution of the Laplacian and to Poisson's equation for gravitational potential. Although it is related to gravity, it is not the main mathematical reason why the potential energy change is independent of path. Therefore, the answer is $\boxed{(3)}$.

□

Problem 2020.5. Among the following mathematics-related academic institutions, which one is actually a degree-granting school?

(1) Institut des Hautes Études Scientifiques, IHES
(2) École normale supérieure
(3) Mathematical Sciences Research Institute, MSRI
(4) Institute for Advanced Study, IAS
(5) Collège de France

Solution. École normale supérieure is one of France's most prestigious higher education institutions. It operates educational programs and is naturally regarded as a degree-granting school.

By contrast, IHES, MSRI, and IAS are research institutes rather than degree-granting schools. Collège de France is also a famous academic institution that offers public lectures, but it does not grant degrees.

Therefore, the answer is $\boxed{\text{(2) École normale supérieure}}$.

□

Problem 2021.1. Which is larger, A or B, where

$$A = 1 + \frac{1}{1+\frac{1}{1-\frac{1}{1+4}}}, \qquad B = 1 + \frac{1}{1+\frac{1}{1+\frac{1}{1+3}}}?$$

Solution. First, we compute A:

$$A = 1 + \frac{1}{1+\frac{1}{1-\frac{1}{5}}} = 1 + \frac{1}{1+\frac{1}{\frac{4}{5}}} = 1 + \frac{1}{1+\frac{5}{4}} = 1 + \frac{1}{\frac{9}{4}} = 1 + \frac{4}{9} = \frac{13}{9}.$$

Next, we compute B:

$$B = 1 + \frac{1}{1+\frac{1}{1+\frac{1}{4}}} = 1 + \frac{1}{1+\frac{1}{\frac{5}{4}}} = 1 + \frac{1}{1+\frac{4}{5}} = 1 + \frac{1}{\frac{9}{5}} = 1 + \frac{5}{9} = \frac{14}{9}.$$

Since

$$\frac{13}{9} < \frac{14}{9},$$

we have $A < B$. Therefore, the answer is $\boxed{B}$.

□

Problem 2021.2. Let

$$A(n) = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^{n+1}.$$

What is the integer closest to $A(7)$?

Solution. Recall Binet's formula for the Fibonacci sequence:

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

Here

$$A(7) = \frac{\varphi^8}{\sqrt{5}}.$$

By Binet's formula,

$$F_8 = \frac{\varphi^8 - \psi^8}{\sqrt{5}}.$$

Since $|\psi| < 1$, the term $\dfrac{\psi^8}{\sqrt{5}}$ is very small. Thus

$$A(7) = \frac{\varphi^8}{\sqrt{5}} = F_8 + \frac{\psi^8}{\sqrt{5}}$$

is very close to F_8.

The Fibonacci numbers are

$$F_0 = 0, \quad F_1 = 1, \quad F_2 = 1, \quad F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \quad F_7 = 13, \quad F_8 = 21.$$

Therefore, the integer closest to $A(7)$ is $\boxed{21}$.

□

Problem 2021.3. Choose all groups among the following that are not finitely generated:

(1) The fundamental group of an open unit disk with finitely many punctures.
(2) $\mathrm{SL}_2(\mathbb{Z})$, the group of 2×2 integer matrices with determinant 1.
(3) $\mathbb{C}^\times$, the group of nonzero complex numbers under multiplication.
(4) The Monster group, the largest sporadic finite simple group.

Solution. First, consider an open unit disk with finitely many punctures. If the removed points are $p_1, \ldots, p_n$, then the fundamental group is isomorphic to the free group on n generators:

$$\pi_1(D \setminus \{p_1, \ldots, p_n\}) \cong F_n.$$

Hence the group in choice (1) is finitely generated.

The group $\mathrm{SL}_2(\mathbb{Z})$ is also finitely generated. For example, it is generated by the two matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Thus choice (2) is finitely generated.

On the other hand, $\mathbb{C}^\times$ is an uncountable abelian group. Every finitely generated group is countable, so $\mathbb{C}^\times$ cannot be finitely generated. Therefore, choice (3) is not finitely generated.

Finally, the Monster group is a finite group. Every finite group is finitely generated, since one may simply take all of its elements as generators. Hence choice (4) is finitely generated.

Therefore, the only group in the list that is not finitely generated is $\boxed{(3)\ \mathbb{C}^\times}$.

□

Problem 2021.4. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Find all eigenvalues of $\mathbf{A}$ together with their multiplicities. Also determine whether $\mathbf{A}$ is diagonalizable.

Solution. We can write $\mathbf{A}$ as

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}.$$

Thus $\mathbf{A}$ is a rank-one matrix.

First, let

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{A}\mathbf{v} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix} = 6\mathbf{v}.$$

Hence 6 is an eigenvalue of $\mathbf{A}$.

Since $\mathbf{A}$ has rank 1, its nullity is 2. Therefore, the eigenspace corresponding to the eigenvalue 0 has dimension 2. Indeed,

$$\mathbf{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y + 3z \\ x + 2y + 3z \\ x + 2y + 3z \end{pmatrix},$$

so

$$E_0 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y + 3z = 0 \right\}.$$

Thus the geometric multiplicity of 0 is 2.

Now

$$\operatorname{tr}(\mathbf{A}) = 1 + 2 + 3 = 6.$$

Since the sum of the eigenvalues, counted with algebraic multiplicity, is equal to the trace, the eigenvalues are

$$6, \quad 0, \quad 0.$$

Hence the algebraic multiplicities are

$$6 : 1, \qquad 0 : 2.$$

Finally, the eigenspace for 6 has dimension 1, and the eigenspace for 0 has dimension 2. The total dimension of the eigenspaces is therefore

$$1 + 2 = 3.$$

Hence there exists a basis of $\mathbb{R}^3$ consisting of eigenvectors of $\mathbf{A}$. Therefore, $\mathbf{A}$ is diagonalizable.

Thus the eigenvalues are $\boxed{6, 0, 0}$, with algebraic multiplicities 1 and 2 respectively, and $\boxed{\mathbf{A} \text{ is diagonalizable}}$.

□

Problem 2022.1. Which of the following numbers is the largest?

(1) e^{π}
(2) π^{e}
(3) 7π
(4) $\binom{7}{2}$
(5) $4!$

Solution. We compare the values directly:

$$e^{\pi} \approx 23.1407, \qquad \pi^{e} \approx 22.4592, \qquad 7\pi \approx 21.9911.$$

Also,

$$\binom{7}{2} = \frac{7 \cdot 6}{2} = 21, \qquad 4! = 24.$$

Therefore, the largest number among the choices is $\boxed{(5)\ 4!}$.

$\square$

Part IV

Expected Problems